

01 Sets (Set Concepts)

Mathematics

Big question: How many stones are in my hand?

Think silently first. Then be ready to explain one idea.

Today's mission

Goal: I can distinguish between finite and infinite sets using practical examples

1

from their environment.

2

You will use: Can We Count Everything?, Defining Our Sets

3

Success: explain, try, and answer in your own words.

Starter: think first

Think: How many stones are in my hand?

1

Pair: compare your first answer with a partner.

2

Share: connect your answer to Can We Count Everything?.

Can We Count Everything?

Key idea: Concrete exploration of countable objects (stones) vs. uncountable

- 1 quantities (sand).
- 2 Practical counting of stones.
- 3 Attempting to count grains of sand.
- 4 Identifying which count can be finished.

Defining Our Sets

Watch the example.

1

Finite Set: Has a countable number of members.

2

Infinite Set: Has members that go on forever.

3

Examples from daily life.

4

Name the step that helped you most.

Try it together

Can you tell me exactly how many grains are in that pile of sand?

1

Which count was easier to finish?

2

What is a finite set?

3

Agree on one answer before writing.

Activity challenge

Group task:

1

In groups of five, learners count a pile of 10 bottle caps and then try to count the

2

number of blades of grass in a small patch of the school compound. They discuss which

3

one has a definite end.

4

Students write the definitions of finite and infinite sets in their...

Exit ticket

Answer: Can a very large set still be finite? (Yes, if it has an end).

1

Use one complete sentence.

2

Include one key word: Set, elements, counting

3

Hand in or read your answer aloud.